

► RENAL—EMBRYOLOGY

Kidney embryology

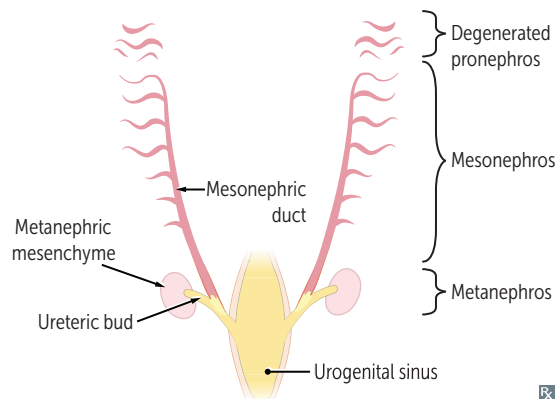
Pronephros—week 4 of development; then degenerates.

Mesonephros—week 4 of development; functions as interim kidney for 1st trimester in both sexes. Mesonephric ducts persist in the male genital system as Wolffian duct, forming ductus deferens and epididymis. Mesonephric ducts degenerate in females.

Metanephros—permanent; first appears in week 5 of development; nephrogenesis is normally completed by week 36 of gestation.

- Ureteric bud (metanephric diverticulum)—fully canalized by week 10 of development; derived from mesonephric duct to form ureters, pelvises, calyces, and collecting ducts
- Metanephric mesenchyme (ie, metanephric blastema)—ureteric bud interacts with this tissue to induce differentiation and formation of glomerulus through distal convoluted tubule (DCT)
- Aberrant interaction between these 2 tissues may result in several congenital malformations of the kidney (eg, renal agenesis, multicystic dysplastic kidney)

Ureteropelvic junction → last part of ureter to canalize; if doesn't fully canalize → congenital obstruction. Can be unilateral or bilateral. Most common pathologic cause of prenatal hydronephrosis. Detected by prenatal ultrasound.

**Potter sequence**

Oligohydramnios → compression of developing fetus → limb deformities (eg, club feet), facial anomalies (eg, low-set ears and retrognathia, flattened nose), compression of chest and lack of amniotic fluid aspiration into fetal lungs → pulmonary hypoplasia (cause of death).

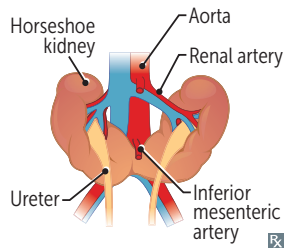
Caused by chronic placental insufficiency or reduced fetal urine output, including ARPKD, obstructive uropathy (eg, posterior urethral valves), bilateral renal agenesis, preterm premature rupture of membranes, maternal ACE inhibitor use.

Babies who can't "Pee" in utero develop **P**otter sequence.

POTTER sequence associated with:

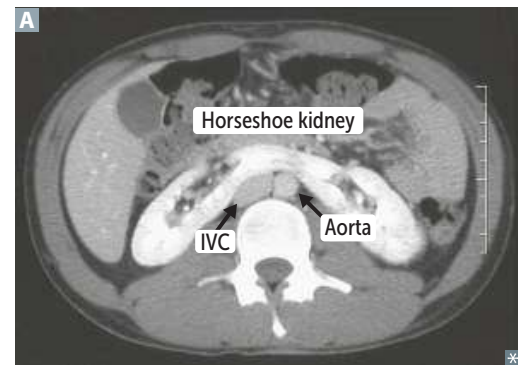
- P**ulmonary hypoplasia
- O**ligohydramnios (trigger)
- T**wisted face
- T**wisted skin
- E**xtrernity defects
- R**enal failure (in utero)

Horseshoe kidney



Inferior poles of both kidneys fuse abnormally **A**. As they ascend from pelvis during fetal development, horseshoe kidneys get trapped under inferior mesenteric artery and remain low in the abdomen. Kidneys can function normally, but associated with hydronephrosis (eg, ureteropelvic junction obstruction), renal stones, infection, ↑ risk of renal cancer.

Higher incidence in chromosomal aneuploidy (eg, Turner syndrome, trisomies 13, 18, 21).



Congenital solitary functioning kidney

Condition of being born with only one functioning kidney. Majority asymptomatic with compensatory hypertrophy of contralateral kidney, but anomalies in contralateral kidney are common. Often diagnosed prenatally via ultrasound bilateral agenesis or dysplasia leads to Potter sequence.

Unilateral renal agenesis

Ureteric bud fails to develop and induce differentiation of metanephric mesenchyme → complete absence of kidney and ureter.

Multicystic dysplastic kidney

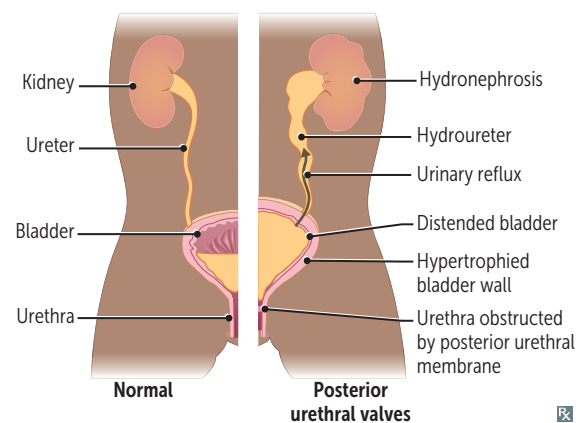
Ureteric bud develops, but fails to induce differentiation of metanephric mesenchyme → nonfunctional kidney consisting of cysts and connective tissue. Nongenetic inheritance, though tends to run in families; usually unilateral.

Duplex collecting system

Bifurcation of ureteric bud before it enters the metanephric blastema creates a Y-shaped bifid ureter. Duplex collecting system can alternatively occur through two ureteric buds reaching and interacting with metanephric blastema. Strongly associated with vesicoureteral reflux and/or ureteral obstruction, ↑ risk for UTIs. Frequently presents with hydronephrosis.

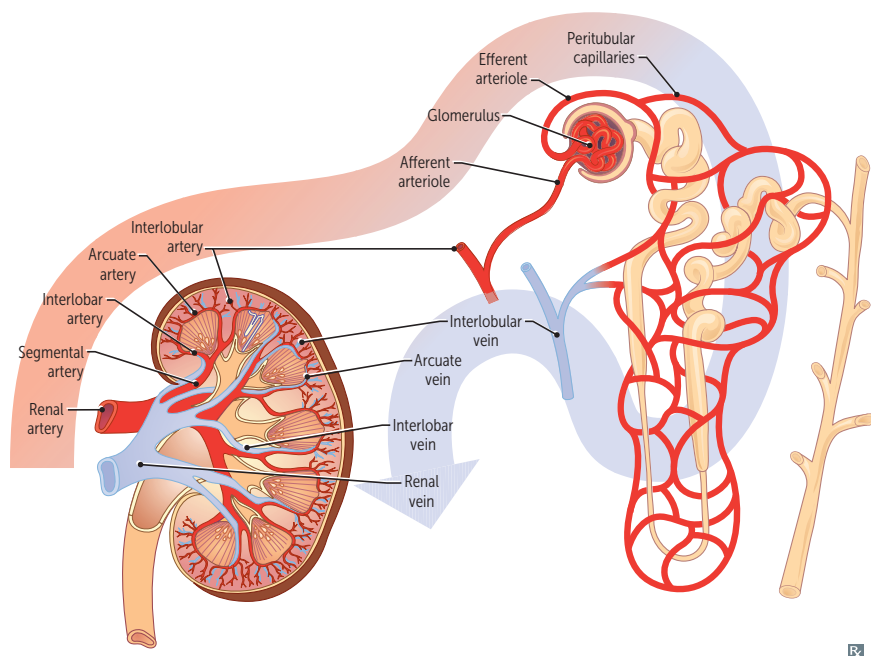
Posterior urethral valves

Membrane remnant in posterior (prostatic) urethra in males; its persistence can lead to urethral obstruction. Diagnosed prenatally by bilateral hydronephrosis and dilated or thick-walled bladder on ultrasound. Severe obstruction in fetus associated with oligohydramnios. Most common cause of bladder outlet obstruction in male infants.



Vesicoureteral reflux

Retrograde flow of urine from bladder toward upper urinary tract. Can be 1° due to abnormal/insufficient insertion of the ureter within the vesicular wall (ureterovesical junction [UVJ]) or 2° due to abnormally high bladder pressure resulting in retrograde flow via the UVJ. ↑ risk of recurrent UTIs.

► RENAL—ANATOMY**Renal blood flow**

Left renal vein receives two additional veins: left suprarenal and left gonadal veins.

Renal medulla receives significantly less blood flow than the renal cortex. This makes medulla very sensitive to hypoxia and vulnerable to ischemic damage (eg, ATN).

Left kidney is most commonly taken during Living donor transplant because it has a Longer renal vein (rule of Ls).

Glomerular anatomy